

Shor's Algorithm without Qubit Registers — the Photonic Perspective

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.1 Preparation and Transformation: Two Separate Processes

Summary

Shor's algorithm falls into two processes that are regularly conflated although they have nothing to do with each other:

	Preparation	Transformation
Input	numbers a, N	prepared signal
Output	signal with $f(x) = a^x \bmod N$	frequency representation
Nature	classical arithmetic	wave operation
Effort	expensive ($O(n^3)$ gates)	cheap ($O(n^2)$)

Preparation is difficult because modular exponentiation is difficult – equally so in every technology. It does *not* depend on quantum-mechanical properties. The transformation is the same wave operation in every technology and cheap in each. The difference between optics and a qubit register lies in neither, but in operating temperature and coupling mechanism.

Three separate parameter spaces

To avoid confusion this document distinguishes strictly:

1. **Geometric space:** $\xi = 4/30000 \approx 1.333 \times 10^{-4}$ – the structural parameter of FFGFT. Fixed, not adjustable.
2. **Algebraic space:** $\xi_{\text{Higgs}} \approx 1.038 \times 10^{-5}$ – field correction in the vacuum expectation value.
3. **Number-theoretic space** $\mathbb{Z}/N\mathbb{Z}$: *no* universal parameter. The period r is determined by $\text{lcm}(p-1, q-1)$ – an individual arithmetic property of N . Factorisation lives exclusively in the third space. Neither ξ nor ξ_{Higgs} has a function there.

.2 Preparation

Input: the numbers a and N . **Output:** a state or signal carrying the sequence $f(x) = a^x \bmod N$.

What happens here

[K] Modular exponentiation is a *classical, deterministic* function. Its evaluation contains no quantum-mechanical component: on a quantum computer it is executed as a reversible classical circuit, in an optical setup it is imprinted as a spatial or spectral structure.

The cost

[K] The effort depends on the arithmetic, not on the chosen technology:

Realisation	Effort	for RSA-2048
Quantum circuit	$O(n^3)$ gates	$8.6 \cdot 10^9$ gates
Optical mask	2^n elements	10^{617} elements
Classical table	2^n values	10^{617} values

The quantum circuit is the cheapest known variant: it encodes the computational rule rather than the values and therefore requires only polynomial effort. An optical mask would have to carry every function value individually – for RSA-2048 more elements than the observable universe has atoms (10^{80}).

Preparation is an arithmetic problem

It is expensive because modular exponentiation is expensive – and that holds in every technology. A method promising to bypass it shifts the effort rather than removing it.
Preparation does not depend on quantum-mechanical properties.

What superposition contributes here

[X] Nothing. The state $\sum_x |x\rangle |a^x \bmod N\rangle$ does contain all pairs, but a measurement yields exactly one – drawn at random. Afterwards one knows as much as after a single classical evaluation. Superposition does not simplify the arithmetic; it applies the same circuit across all inputs.

.3 The Hybrid Preparation: Binary Control Meets Analogue Wave

Pure binary implementation of modular exponentiation drowns in $O(n^3)$ gates. Purely analogue implementation fails at thermal noise and the modulo problem. A hybrid approach exploits the strengths of both:

- **The binary part (the control):** Exact, noise-free discretisation of the modulo logic – governs which phase paths are enabled.
- **The analogue part (the wave):** Continuous, parallel phase accumulation via interference and mixer structures in real time.

What happens physically

The analogue wave performs multiplication via *coherent phase superposition*. Mixing two waves with phases ϕ_1 and ϕ_2 produces:

$$\Psi_{\text{total}} \propto e^{i\phi_1} + e^{i\phi_2}$$

or for coupled systems $|\phi_1\rangle \otimes |\phi_2\rangle$. In classical engineering this is coherent summation or frequency mixing; in quantum physics the same mathematical structure is called superposition.

Why physics introduces new terms here

1. **Discretisation by single quanta:** At the single-photon level the analogue wave becomes a probability distribution – analogue superposition becomes quantum superposition.
2. **Escaping the noise problem:** Classical analogue mixers suffer from thermal noise ($k \cdot T \cdot B$). Operating in the single-photon regime enables noise-free switching via quantum error correction.

Conclusion on hybrid preparation

Hybrid preparation = binary-controlled interference network

The hybrid preparation is a *phased-array-type interference network* that is binary-conditioned. The “superposition” here is the physical description of analogue wave superposition – not a quantum- mechanical magic.

Cost of hybrid preparation

Implementation	Gate count	Characteristic
Purely binary	$O(n^3)$	Drowns for large n
Purely analogue	—	Fails at noise
Hybrid	Reduced	Phased-array interference

.4 The Fourier Transform

Input: the prepared signal. **Output:** its frequency representation.

What happens here

[K] The Fourier transform is a *wave operation*. It takes no interest in where the signal comes from or what it means – it translates one representation into another. No new information arises: what is absent from the input signal does not appear in the output.

The cost

[K] It is cheap in every realisation:

Realisation	Effort	Remark
Quantum Fourier transform	$O(n^2)$ gates	sequential
Optical MZI network	transit time	simultaneous, speed of light
Classical FFT	$O(2^n \cdot n)$	on explicit values

In the quantum register and in the optical network the effort is small. The difference between the two lies in operating temperature and coupling mechanism (Section .5), not in the transform itself.

What it is for

The period r sought is a *global* property of the sequence: no single function value carries it; it becomes visible only by comparing several values. The Fourier transform performs this comparison physically – interference cancels everything except the frequencies k/r , and from a few measurements r follows by continued-fraction expansion.

Two separate processes

Preparation: arithmetic, expensive, difficult independently of technology. *Transformation:* wave-mechanical, cheap, the same operation in every technology. The two are unconnected. Preparation can be improved without touching the transform, and vice versa.

.5 Photonic Realisation

MZI networks

A Mach-Zehnder interferometer realises phase rotations

$$\Delta\phi = \frac{2\pi jk}{n}, \quad (1)$$

which correspond exactly to the twiddle factors of the discrete Fourier transform. A network of such interferometers performs the Fourier transform as a direct wave operation.

[B] The Fourier transform is therefore not a distinguishing feature of the quantum computer. It is a wave operation realisable optically, acoustically, or electronically. The difference lies in state preparation, not in the transform.

The thermal advantage of optical realisation

[K] Whether a mode sits in its ground state without cooling is decided by the ratio $h\nu/(k_B T)$. The mean thermal occupation is

$$\bar{n} = \frac{1}{e^{h\nu/k_B T} - 1}. \quad (2)$$

System	ν (Hz)	T (K)	$h\nu/k_B T$	$\bar{n}$
Superconducting qubit	$5.0 \cdot 10^9$	300	$8.0 \cdot 10^{-4}$	$1.25 \cdot 10^3$
Superconducting qubit	$5.0 \cdot 10^9$	0.020	12.0	$6.2 \cdot 10^{-6}$
Photon 1550 nm	$1.93 \cdot 10^{14}$	300	30.9	$3.7 \cdot 10^{-14}$
Photon 600 nm	$5.0 \cdot 10^{14}$	300	80.0	$1.8 \cdot 10^{-35}$

At 5 GHz and room temperature some **1250 thermal excitations** sit in the mode – the system is fully thermalised. For an occupation below 0.01, $T < 52$ mK is required. At optical frequencies the ground state is occupied essentially exactly at room temperature.

Cooling cost. The Carnot limit demands at least $3 \cdot 10^4$ units of work per unit of heat removed at 10 mK; in practice considerably more. A dilution refrigerator draws 10–25 kW continuously; an optical network draws none.

Thermodynamically privileged, not computationally

The thermal advantage follows directly from $h\nu/k_B T$ and is the strongest known advantage of photonic over superconducting realisation. It concerns *infrastructure* – cooling, energy, size – not computational power. Both realisations can in principle run the same algorithms; photonics has difficulties of its own (deterministic two-photon gates, photon loss as the dominant error channel).

Where the exponential dimension comes from

[K] An opposition of “optics versus quantum computer” would be inaccurate – it would set a technology against a computational model. Photonic quantum computers exist (KLM scheme, measurement-based approaches). What matters is solely the structure of the state space:

Arrangement	Dimension
classical wave in n modes	n
one photon in n modes	n (the same)
n entangled two-level systems	2^n
m indistinguishable photons in n modes	$\binom{n+m-1}{m}$

A single photon in an MZI network realises the same n -dimensional unitary as a classical wave – classically simulable. With several indistinguishable photons the dimension grows exponentially: 25 photons in 50 modes give $3.5 \cdot 10^{19}$. The jump comes from the *tensor product structure*, not from the carrier medium.

What carries the computational advantage – an open question

[K] It would be natural to attribute the advantage to entanglement. The Gottesman-Knill theorem (1998) refutes this: any circuit consisting solely of Clifford gates (H , S , CNOT) is classically simulable in polynomial time, regardless of the entanglement produced. The GHZ state $(|0 \dots 0\rangle + |1 \dots 1\rangle)/\sqrt{2}$ is maximally entangled and arises from H and $n - 1$ CNOTs – at 1000 qubits some $2 \cdot 10^6$ bits suffice for the complete classical description.

Further counterexamples: DQC1 shows a conjectured advantage with almost no entanglement; boson sampling rests on indistinguishability and permanents. Jozsa and Linden (2003) show entanglement to be *necessary* for exponential advantage with pure states – about sufficiency the theorem says nothing.

[X] Which quantity carries the computational advantage is not settled. Contextuality, non-stabiliserness (“magic”) and the interference structure itself are discussed; none of these candidates is established as the sole carrier. This document names the question as open rather than answering it.

What follows for factorisation. The optical realisation does not solve the problem: the limiting step is state preparation – the evaluation of $a^x \bmod N$ over all x – and that is arithmetic. A network can transform a prepared superposition; it cannot create one.

Topological constraint

[K] For a wave in phase space to form a closed resonance, the *logarithmic winding number* $W = \log_2(r)$ must be rational – not the frequency ratio r itself:

Case	Ratio r	Winding number W	Orbit
Pure fifth	$3/2$ (rational)	0.584963 (irrational)	open
Tempered fifth	$2^{7/12}$ (irrational)	$7/12$ (rational)	closed

Undamped oscillations on pure numerical ratios produce irrational winding numbers and diverge as an open spiral. An optical network therefore requires a discrete, rationalised phase grid to achieve sharp resonance points – closure is an artefact of discretisation.

.6 Structural Limits

The Weyl obstruction

[K] A quantum computer with n qubits spans a compact, finite state space. By Weyl's law its eigenvalue density grows by a power law:

$$N(\lambda) \sim C_d \cdot \lambda^{d/2}. \quad (3)$$

Arithmetic counting densities, by contrast, grow as

$$N(T) \sim T \ln T \quad (4)$$

(Riemann-von Mangoldt). $T \ln T$ is not a power.

Consequence: a finite state space cannot in principle carry such a spectrum in full – it carries at most an initial segment. Infinite spectra of this density would require an *unrolled dilation domain* (Docs. 315, 316), not a compact resonance space.

Important delimitation

This obstruction does **not** apply to Shor's algorithm. Shor seeks *one* period of *one* concrete number N – a finite task for which a register of $2n + 3$ qubits suffices. The Weyl obstruction concerns the complete representation of *infinite* arithmetic spectra. Invoking it against Shor overextends it.

What actually limits Shor at large N is the gate count of modular exponentiation (growing as n^3) and the error-correction overhead (roughly 10^3 physical per logical qubit). For RSA-2048: about 4099 logical qubits, 8.6×10^9 gates, 4.1×10^6 physical qubits. That is a *resource limit* – in principle surmountable by better hardware, unlike a structural one.

Primes as generators

[K] The load-bearing resonance structure rests on primes. The bicoherence test on 2469 Riemann zeros (Doc. 316) shows phase coupling exactly at prime powers ($b^2 = 0.76\text{--}0.85$) and none at composite numbers ($b^2 = 0.018\text{--}0.025$).

The physical counterpart is the harmonic series: in $n f_0$, exactly the prime orders introduce new tonal qualities; $4 = 2 \cdot 2$, $6 = 2 \cdot 3$, $9 = 3 \cdot 3$ are combinations of axes already present.

.7 Placing the Quantum Computer

[K] Shor's algorithm factorises on quantum hardware in theoretically $O((\log N)^3)$ – a real achievement for a specific problem. It does not follow that:

- **[X]** *Quantum computers hold a general exponential advantage.* This is not established for the vast majority of problems. Shor is a single case, not evidence of a general advantage.
- **[X]** *The advantage lies in the Fourier transform itself.* That is a wave operation and optically realisable. The bottleneck is state preparation.
- **[X]** *Quantum computers contribute to structural questions of number theory.* The Weyl obstruction applies to finite state spaces without qualification.

The core: reading out a period is a spectroscopic measurement problem on prepared wave fields. The measurement finds what was prepared; the preparation is the expensive part.

.8 Balance

Statement	Status
QFT is an interference operation	[K]
QFT generates no new information	[K] measured
Winding number decides closure	[K]
Primes carry the resonance structure	[K] measured by bicoherence
Fourier transform optically realisable	[B]
Optics thermodynamically privileged	[K] $h\nu/k_B T$
Exponential dimension from the medium	[X] tensor structure
Entanglement carries the advantage	[X] Gottesman-Knill
Preparation is arithmetic, not quantum-mechanical	[K]
Preparation and transformation are independent	[K]
General quantum advantage	[X] not established
Finite space carries infinite $T \ln T$ spectrum	[X] Weyl
Weyl obstruction limits Shor	[X] resource limit
ξ has a function in $\mathbb{Z}/N\mathbb{Z}$	[X] separate spaces

The load-bearing sentence. The Fourier transform is a projection – it finds structure but does not generate it. This holds equally for the classical FFT, the optical realisation, and the QFT. Factorisation lives in the space $\mathbb{Z}/N\mathbb{Z}$, where the geometry parameters of FFGFT have no function.

.9 Relation to the Corpus

The findings used are from Doc. 316 (August 2026): Weyl obstruction, bicoherence test, winding number versus frequency ratio, and the Fourier transform as a projection. The revision is declared under R75 in Doc. 190.

.10 Verification

The statements of this document are reproducible by two scripts (standard library only, target values as assertions):

- `s1_periodensuche.py` – correctness of period finding, scaling measurement (measured growth exponent of the mean order 0.95, hence effectively $O(N)$), prime factorisation of the σ denominators, comparison of methods.
- `s2_grenzen.py` – Weyl obstruction against an equidistant register, delimitation from Shor's task, resource estimate, winding-number check.
- `s3_thermik_zustandsraum.py` – thermal occupation by carrier frequency, Carnot cooling cost, dimension comparison of state spaces, Gottesman-Knill.